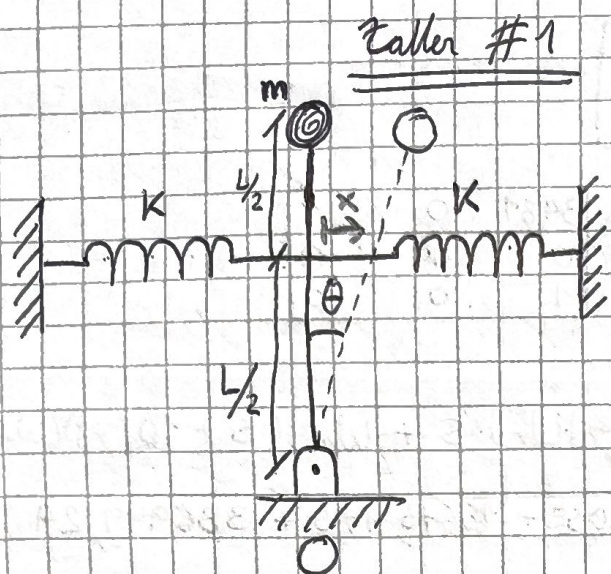




$$T = \frac{1}{2} m (L\dot{\theta})^2$$

$$T = \frac{1}{2} m L^2 \dot{\theta}^2$$

$$V_g = mgL(1 - \cos\theta)$$

$$V_g = \frac{1}{2} mgL\theta^2$$

$$x = \frac{L}{2} \sin\theta \approx \frac{L}{2} \theta$$

$$V_k = 2 \cdot \frac{1}{2} K x^2 = K x^2$$

$$= K \left(\frac{L}{2} \theta \right)^2 = \frac{KL^2}{4} \theta^2$$

$$V = \frac{1}{2} mgL\theta^2 + \frac{KL^2}{4} \theta^2$$

$$T - V = mL^2 \ddot{\theta} + \left(mgL + \frac{KL^2}{2} \right) \theta = \tau(t)$$

$$\ddot{\theta} + \left(\frac{g}{L} + \frac{K}{2m} \right) \theta = \frac{1}{mL^2} \tau(t)$$

$$x_1 = \theta$$

$$x_2 = \dot{\theta}$$

$$\dot{x}_1 = x_2$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\left(\frac{g}{L} + \frac{K}{2m}\right) & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{mL^2} \end{bmatrix} \tau(t)$$

$$m = 1 \text{ kg}$$

$$L = 1 \text{ m}$$

$$K = 50 \text{ N/m}$$

$$g = 9,806 \text{ m/s}^2$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} \tau(t)$$

$$\text{POLE} \begin{bmatrix} 0 & 1 \\ -34,806 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1$$

$$\det(sI - A) = s^2 + 34,806$$

$$t_f = \frac{1}{s^2 + 34,806}$$

$$t_{sla} = 0,2 \text{ s}$$

$$t_{sd} = 1,5 \text{ s}$$

$$\zeta = 0,7$$

$$\omega_n = \frac{4}{0,7 \cdot 1,5} = 3809$$

$$S = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \det(S) = -1; \text{Controllable}$$

$$\hat{A} = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -34,806 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}; \hat{B} = \begin{bmatrix} B \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\hat{S} = [\hat{B} \quad \hat{A}\hat{B} \quad \hat{A}^2\hat{B}] = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -34,81 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\det(SI - \hat{A}) = S^3 + 34,81S \quad ; \quad M = \begin{bmatrix} 34,81 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$T = \hat{S} \cdot M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix} ; \quad P_{old} = (S^2 + 2\zeta_d \omega_n S + \omega_n^2)(S + 10(\zeta_d)(\omega_n))$$

$$= S^3 + 320S^2 + 15673,47S + 386999,24$$

$$K = \begin{bmatrix} (386999,24 - 0) & (15673,47 - 34,81) & (320 - 0) \end{bmatrix} T^{-1}$$

$$K = \begin{bmatrix} 15638,66 & 320 & -386999 \\ K_1 & K_2 & -K_i \end{bmatrix}$$

Le agregó una perturbación

$$E = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Observador

$$C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$\sigma = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -34,81 & 0 \end{bmatrix} ; \quad \det(\sigma) = 34,81$$

$$M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Q = (M\sigma)^{-1} = \begin{bmatrix} -0,02873 & 0 \\ 0 & 1 \end{bmatrix}$$

$$L = Q \begin{bmatrix} -145623,249 \\ 0 \end{bmatrix} = \begin{bmatrix} -4183,85 \\ 0 \end{bmatrix}$$

$$\frac{1}{S + 34,81}$$

$$ess = 0$$

$$tipo = 0 \rightarrow 1I$$

$$Orden 1 \rightarrow 2$$

$$\zeta_d = 0,7 \quad \omega_n = 38,09$$

$$t_{sd} = 1,5s$$

P I

$$PI(s) = K_p + \frac{K_i}{s}$$

$$\frac{K_p S + K_i}{s}$$

$$K_p S + K_i + S^2 + 34,81S = P.O.$$

$$S^2 + (K_p + 34,81)S + K_i = S^2 + 53,326S + 1450,85$$

$$K_p + 34,81 = 53,326$$

$$K_p = 18,516$$

$$K_i = 1450,85$$

$$z = 0,7$$

$$0,25 \leq \omega_n T \leq 1$$

$$0,0066 \leq T \leq 0,0263$$

$$T = 0,01$$

$$s = \frac{1 - z^{-1}}{T}$$

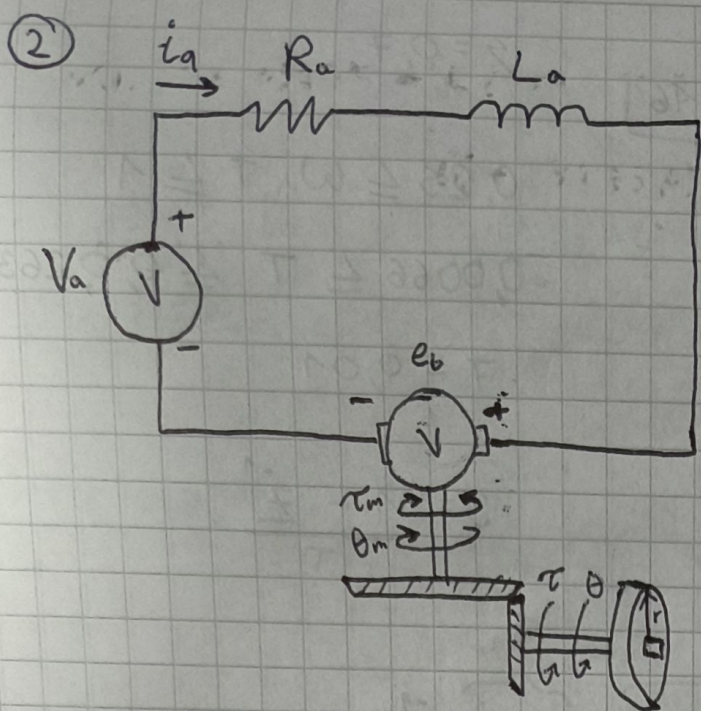
$$\frac{K_p s + K_i}{s} = \frac{K_p \left(\frac{1 - z^{-1}}{0,01} \right) + K_i}{\left(\frac{1 - z^{-1}}{0,01} \right)}$$

$$\frac{K_p - K_p z^{-1} + K_i(0,01)}{1 - z^{-1}} = \frac{(K_p + K_i(0,01)) + (-K_p)z^{-1}}{1 - z^{-1}}$$

$$q_0 = 53,326 + (1450,85)0,01 = 67,835$$

$$q_1 = -1450,85$$

C2D



$$V_a = R_a i_a + L_a \frac{di_a}{dt} + e_b$$

$$e_b = K_e \omega ; \omega = \dot{\theta}$$

$$L_a \dot{i}_a = -R_a i_a - K_e \omega + V_a$$

$$J \ddot{\omega} + B \dot{\omega} = \tau_m \quad X_1 = i_a$$

$$\tau_m = K_t i_a \quad X_2 = \omega$$

$$J \ddot{\omega} = -B \dot{\omega} + K_t i_a \quad X_3 = \theta$$

$$U = V_a$$

$$\dot{X}_3 = X_2$$

$$\begin{bmatrix} \dot{i}_a \\ \dot{\omega} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -\frac{R_a}{L_a} & -\frac{K_e}{L_a} & 0 \\ \frac{K_t}{J} & -\frac{B}{J} & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} i_a \\ \omega \\ \theta \end{bmatrix} + \begin{bmatrix} \frac{1}{L_a} \\ 0 \\ 0 \end{bmatrix} V_a$$

$$y = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_a \\ \omega \\ \theta \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} V_a$$

$$R_a = 2 \Omega ; L_a = 0,5 H$$

$$K_e = 0,1 \frac{Vs}{rad} ; K_t = 0,1 \frac{N \cdot m}{A}$$

$$J = 0,02 \text{ Kg} \cdot \text{m}^2 ; B = 0,02 \frac{N \cdot m \cdot s}{rad}$$

$$\begin{bmatrix} -4 & -0,2 & 0 \\ 5 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$S = \begin{bmatrix} B & AB & A^2 B \end{bmatrix} = \begin{bmatrix} 2 & -8 & 30 \\ 0 & 10 & -50 \\ 0 & 0 & 10 \end{bmatrix}$$

$$\det(S) = 200$$

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix}$$

$$\det(SI - A) = s^3 + 5s^2 + 5s$$

$$TF_{\theta} = \frac{10}{s^3 + 5s^2 + 5s}$$

$$t_{sla} \approx 2s ; t_{sd} = 1,75s ; \zeta_d = 0,85$$

$$\omega_{md} = 2,689$$

$$A^* = \begin{bmatrix} A - BK & BK_{i_2} & BK_{i_1} \\ 0 & 0 & 1 \\ -C & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -K_1 & -K_2 - 5 & -K_3 - 5 & K_{i_2} & K_{i_1} \\ 0 & 0 & 0 & 0 & 1 \\ -10 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\det(sI - A^*) = s^5 + (K_3 + 5)s^4 + (K_2 + 5)s^3 + (K_1)s^2 + 10K_i s + 10K_{i2}$$

$$P_o(d) = (s^2 + 2(0,85)(2,689)s + 2,689)(s + 10(0,85)(2,689))^3$$

$$P_o(d) = s^5 + 73,14 s^4 + 1888,05 s^3 + 19602,55 s^2 + 65924,27 s + 86351,89$$

$$K_3 + 5 = 73,14$$

$$K_3 = 68,14$$

$$K_2 + 5 = 1888,05$$

$$K_2 = 1883,05$$

$$K_1 = 19602,55$$

$$10K_i = 65924,27$$

$$K_i = 6592,427$$

$$10K_{i2} = 86351,89$$

$$K_{i2} = 8635,189$$

$$\sigma = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -4 & -2 & 0 \\ 15 & 1 & 0 \end{bmatrix}; \det(\sigma) = 0 \rightarrow \text{No Observable}$$

$$\frac{10}{s^3 + 5s^2 + 5s}$$

$$ess = 0 \quad \swarrow$$

$$z_d = 0,85 \quad t_{sd} = 1,75 \quad W_{hd} = 2,689$$

$$\text{tipo } 1 \rightarrow 2$$

$$D^2 \quad 0 \quad P \quad I$$

$$\text{Orden } 3 \rightarrow 4$$

$$K_p + \frac{K_i}{s} + K_{d1}s + K_{d2}s^2 = \frac{K_{d2}s^3 + K_{d1}s^2 + K_p s + K_i}{s}$$

$$10K_{d2}s^3 + 10K_{d1}s^2 + 10K_p s + 10K_i + s^4 + 5s^3 + 5s^2 = P_o(d)$$

$$s^4 + (10K_{d2} + 5)s^3 + (10K_{d1} + 5)s^2 + 10K_p s + 10K_i = P_o(d)$$

$$P_o(d) = (s^2 + 2(0,85)(2,689)s + 2,689)(s + 10(0,85)(2,689))^2$$

$$0,74 \leq \zeta \leq 1$$

$$s^4 + 50,28 s^3 + 738,6 s^2 + 2719 s + 3777$$

$$10K_{d2} + 5 = 50,28$$

$$K_{d2} = 4,528$$

$$10K_{d1} + 5 = 738,6$$

$$K_{d1} = 73,36$$

$$10K_p = 2719$$

$$K_p = 271,9$$

$$10K_i = 3777$$

$$K_i = 377,7$$

$$0,25 \leq W_n T \leq 1,5$$

$$0,0929 \leq T \leq 0,558$$

$$T = 0,15$$

$$\frac{Kd_2 s^3 + Kd_1 s^2 + Kp s + Ki}{s} = \frac{Kd_2 \left(\frac{1-\bar{z}^{-1}}{0,15} \right)^3 + Kd_1 \left(\frac{1-\bar{z}^{-1}}{0,15} \right)^2 + Kp \left(\frac{1-\bar{z}^{-1}}{0,15} \right) + Ki}{\left(\frac{1-\bar{z}^{-1}}{0,15} \right)}$$

$$s = \frac{1-\bar{z}^{-1}}{0,15}$$

$$\frac{Kd_2}{0,15^3} \left(1 + 3\bar{z}^{-1} - 3\bar{z}^{-2} + \bar{z}^{-3} \right) + \frac{Kd_1}{0,15^2} \left(1 - 2\bar{z}^{-1} + \bar{z}^{-2} \right) + Kp - Kp\bar{z}^{-1} + 0,15Ki$$

$$\underbrace{\left(\frac{Kd_2}{0,15^3} + \frac{Kd_1}{0,15} + 0,15Ki \right)}_{q_0} + \underbrace{\left(\frac{3Kd_2}{0,15^2} + \frac{2Kd_1}{0,15} - Kp \right)}_{q_1} \bar{z}^{-1} + \underbrace{\left(\frac{Kd_2}{0,15^2} + \frac{Kd_1}{0,15} \right)}_{q_2} \bar{z}^{-2} + \underbrace{\frac{Kd_2}{0,15^3}}_{q_3} \bar{z}^{-3} = \frac{(1-\bar{z}^{-1})^2}{1-2\bar{z}^{-1}+\bar{z}^{-2}} \cdot \frac{(1-\bar{z}^{-1})^3}{(1-\bar{z}^{-1})^3}$$

$$q_0 = 746,966$$

$$q_1 = -646,3$$

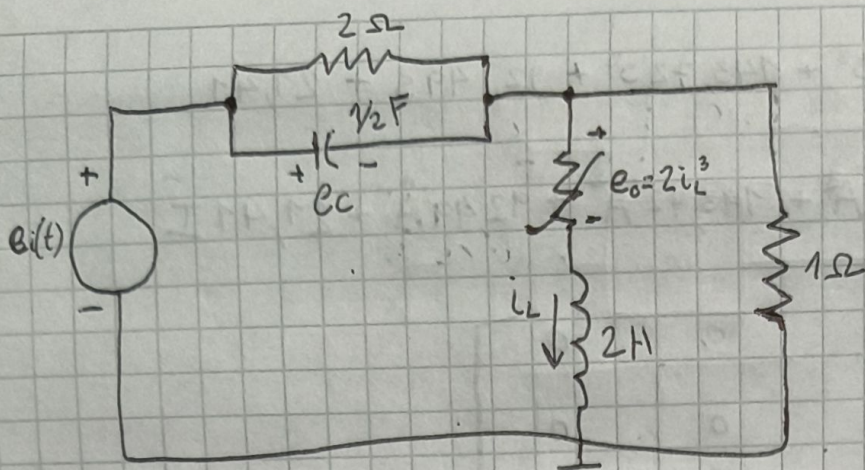
$$q_2 = 690,31$$

$$q_3 = 201,24$$

$$1 + 3(1)^2(-\bar{z}^{-1}) - 3(1)(\bar{z}^{-1})^2 + \bar{z}^{-3}$$

$$1 + 3\bar{z}^{-1} - 3\bar{z}^{-2} + \bar{z}^{-3}$$

~~Dependent~~



$$X_1 = V_c$$

$$X_2 = i_L$$

$$U = e_i$$

$$Y = e_o$$

$$\dot{V}_c = 2\dot{i}_L + 2e_i - 3V_c$$

$$V = e_o + e_n \rightarrow e_o = 2i_L^3$$

$$e_n = V_L = -2\dot{i}_L \rightarrow V = 2i_L^3 + 2\dot{i}_L \quad V = e_i - V_c$$

$$2i_L^3 + 2\dot{i}_L = e_i - V_c \rightarrow \dot{i}_L = \frac{1}{2}e_i - \frac{1}{2}V_c - i_L^3$$

$$\dot{V}_c = -3V_c + 2\dot{i}_L + 2e_i$$

$$X_1 \approx 0$$

$$X_2 \approx 0$$

$$u = 0$$

$$\begin{bmatrix} \dot{V}_c \\ \ddot{i}_L \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} V_c \\ i_L \end{bmatrix} + \begin{bmatrix} 2 \\ \frac{1}{2} \end{bmatrix} e_i$$

$$t_f(s) = \frac{2s + 1}{s^2 + 3s + 1}$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} V_c \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} e_i$$

$$t_{sla} = 10s \rightarrow t_{sd} = 9s$$

$$\zeta d = 0.9 \quad Wnd = \frac{4}{10 \cdot 0.9} = 0.444$$

$$K = [0 \ 0 \ 0 \ 0 \ 1] \hat{S}^{-1} \phi A$$

$$\hat{S} = [\hat{B} \ \hat{A}\hat{B} \ \hat{A}^2\hat{B} \ \hat{A}^3\hat{B} \ \hat{A}^4\hat{B}]$$

$$\hat{A} = \begin{bmatrix} A & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -C & 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -3 & 1 & 0 & 0 & 0 \\ -0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{S} = \begin{bmatrix} 2 & -5 & 13 & -34 & 89 \\ 0.5 & -1 & 2.5 & -6.5 & 17 \\ 0 & 0 & 0 & -2 & 5 \\ 0 & 0 & -2 & 5 & -13 \\ 0 & -2 & 5 & -13 & 34 \end{bmatrix}$$

$$\hat{B} = \begin{bmatrix} B \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0.5 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$Pol D = (s^2 + 2(0.9)(0.44)s + 0.44^2)(s + 10(0.9)(0.44))^3$$

$$P_0(D) = S^5 + 14,22S^4 + 71,35S^3 + 143,72S^2 + 92,49S + 21,41$$

$$\Phi A = [\hat{A}^5 + 14,22\hat{A}^4 + 71,35\hat{A}^3 + 143,72\hat{A}^2 + 92,49\hat{A} + 21,41I]$$

$$\Phi A = \begin{bmatrix} 33,456 & -22,985 & 0 & 0 & 0 \\ 5,7462 & -1,0215 & 0 & 0 & 0 \\ -36,688 & -22,444 & 21,40 & 92,48 & 143,71 \\ -22,43 & -73,37 & 0 & 21,41 & 92,48 \\ 11,49 & -44,86 & 0 & 0 & 21,41 \end{bmatrix}$$

$$K = [29,625 \quad -96,05 \quad -21,41 \quad -49,67 \quad -44,38]$$

Observer $C = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} V_c \\ i_L \end{bmatrix}$

$$\sigma = \begin{bmatrix} \lambda_1 & \lambda_2 \end{bmatrix} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$$

$$\det(\sigma) = 2$$

$$M = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Q = (M \cdot \sigma)^{-1} = \begin{bmatrix} 0 & 1 \\ 0,5 & 0 \end{bmatrix}$$

$$L = Q \begin{bmatrix} 32,28 \\ -3 \end{bmatrix} = \begin{bmatrix} -3 \\ -16,13 \end{bmatrix}$$

$$F_t(s) = \frac{2s+1}{s^2+3s+1} \left(\frac{1}{2s+1} \right)_{\text{ess}} = 0$$

typo 0 → 3
Ordern 2 → 5

$$3d = 0,9 \quad t_{sd} = 9s$$

$$W_{md} = 0,444$$

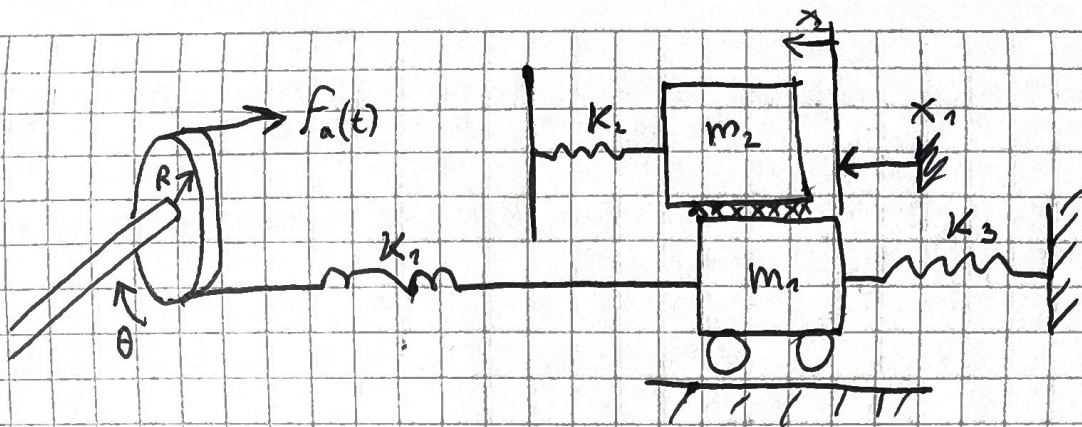
$$D \quad P \quad I \quad I^2 \quad I^3$$

$$K_d s + K_p + \frac{K_{i1}}{s} + \frac{K_{i2}}{s^2} + \frac{K_{i3}}{s^3} = \frac{K_d s^4 + K_p s^3 + K_{i1} s^2 + K_{i2} s + K_{i3}}{s^3}$$

$$S^5 + (K_d + 3)S^4 + (K_p)S^3 + K_{i1}S^2 + K_{i2}S + K_{i3} = P.D.$$

$$S^5 + 12,79S^4 + 57,68S^3 + 104,5S^2 + 60,44S + 12,58 ; K_d = 9,79,$$

$$K_p = 57,68 ; K_{i1} = 104,5 ; K_{i2} = 60,44 ; K_{i3} = 12,58$$



$$m_1 \ddot{x}_1 = f_a(t) - K_3 x_1 + B(\dot{x}_2 - \dot{x}_1) \Rightarrow m_1 \ddot{x}_1 = f_a(t) - K_3 x_1 + B\dot{x}_2 - B\dot{x}_1$$

$$m_2 \ddot{x}_2 = -K_2 x_2 + B(\dot{x}_1 - \dot{x}_2) \Rightarrow m_2 \ddot{x}_2 = -K_2 x_2 + B\dot{x}_1 - B\dot{x}_2$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{K_3}{m_1} & -\frac{B}{m_1} & 0 & \frac{B}{m_1} \\ 0 & 0 & 0 & 0 \\ 0 & \frac{B}{m_2} & -\frac{K_2}{m_2} & -\frac{B}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m_1} f_a(t) \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = x_1$$

$$\dot{x}_1 = x_2$$

$$x_2 = x_3$$

$$\dot{x}_2 = x_4$$

$$C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} f_a(t)$$

$$m_1 = 5 \text{ Kg}; m_2 = 2 \text{ Kg}$$

$$K_2 = 800 \text{ N/m}; K_3 = 500 \text{ N/m}$$

$$B = 40 \text{ N.s/m}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -100 & -8 & 0 & 8 \\ 0 & 0 & 0 & 1 \\ 0 & 20 & -400 & -20 \end{bmatrix} \begin{bmatrix} 0 \\ 0,2 \\ 0 \\ 0 \end{bmatrix}$$

$$t_f(s) = \frac{45}{s^4 + 28s^3 + 500s^2 + 5200s + 40000}$$

$$t_{sla} = 3s \rightarrow t_{sd} = 2,7 \quad \zeta d = 0,7$$

$$\omega_n = \frac{4}{2,7(0,7)} = 2,776$$

$$\hat{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -100 & -8 & 0 & 8 \\ 0 & 0 & 0 & 1 \\ 0 & 20 & -400 & -20 \end{bmatrix}; \hat{B} = \begin{bmatrix} 0 \\ 0,2 \\ 0 \\ 0 \end{bmatrix}$$

$$\hat{S} = \begin{bmatrix} \hat{B} & \hat{A}\hat{B} & \hat{A}^2\hat{B} & \hat{A}^3\hat{B} & \hat{A}^4\hat{B} \end{bmatrix} = \begin{bmatrix} 0 & 0,2 & -1,6 & 24,8 & -434,4 \\ 0,2 & -1,6 & 24,8 & -434,4 & 14080,0 \\ 0 & 0 & 4 & -112 & 1136 \\ 0 & 4 & -112 & 1136 & 3392 \\ 0 & 0 & 0 & -4 & 112 \end{bmatrix}$$

$$\det(sI - \hat{A}) = s^5 + 28s^4 + 500s^3 + 5200s^2 + 40000s$$

$$P_{old} = s^5 + 47,41s^4 + 794,60s^3 + 5401,53s^2 + 12583,423s + 14564,15$$

$$M = \begin{bmatrix} 40000 & 5200 & 500 & 28 & 1 \\ 5200 & 500 & 28 & 1 & 0 \\ 500 & 28 & 1 & 0 & 0 \\ 28 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}; T = \hat{S}M = \begin{bmatrix} 0 & 80 & 4 & 0,2 & 0 \\ 0 & 0 & 80 & 4 & 0,2 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & -4 & 0 & 0 & 0 \end{bmatrix}$$

$$K = \begin{bmatrix} 14564,15 - 0 & 12583,423 - 40000 & 5401,53 - 5200 & 794,6 - 500 & 47,41 - 28 \end{bmatrix} T^{-1}$$

$$K = [-1,58e^{13} \quad 67,73 \quad 1,588e^{13} \quad 7,94e^{11} \quad -3,147e^{14}]$$

$$\begin{array}{l} \frac{48}{s^4 + 28s^3 + 500s^2 + 5200s + 40000} \quad \frac{1}{s} \quad \begin{array}{l} e_{ss} = 0 \quad \checkmark \\ 2I \\ \zeta_{po} = -1 \rightarrow 1 \\ \sigma_{dom} = 4 \rightarrow 6 \end{array} \quad \begin{array}{l} t_{sd} = 0,7 \quad \checkmark \quad \beta = 0,7 \\ W_{nd} = 2,1164 \\ D^3 D^2 D P I I^2 \end{array} \end{array}$$

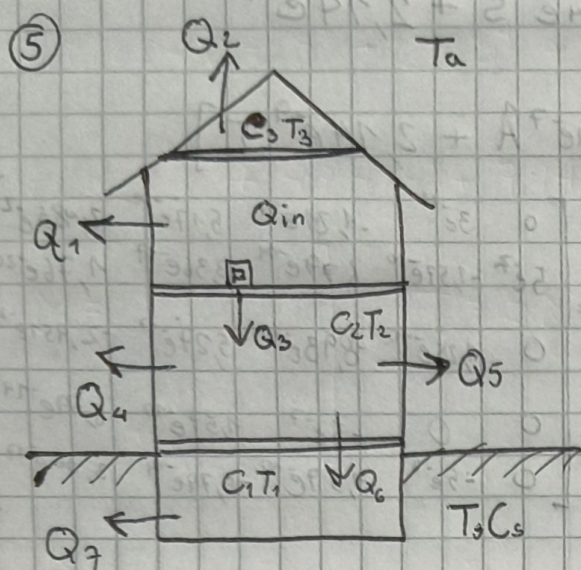
$$K_d s^3 + K_d s^2 + K_d s + K_p + \frac{K_{i1}}{s} + \frac{K_{i2}}{s^2} = \frac{K_d s^5 + K_d s^4 + K_d s^3 + K_p s^2 + K_{i1} s + K_{i2}}{s^2}$$

$$\checkmark \quad \Delta \quad \square \quad * \quad \circ \quad \frac{1}{s^6} + 28s^5 + 500s^4 + 5200s^3 + 40000s^2$$

$$s^6 + (4K_d + 28)s^5 + (4K_d + 500)s^4 + (4K_d + 5200)s^3 + (4K_p + 40000)s^2 + 4K_{i1}s + 4K_{i2}$$

$$P.D. = s^6 + 62,22s^5 + 1497s^4 + 17170s^3 + 92610s^2 + 201000s + 215800$$

$$K_d = 8,555; K_d = 249,25; K_d = 2492,5; K_p = 13759,5; K_{i1} = 50230; K_{i2} = 53950$$



$$X_1 = T_1$$

$$Y = T_2$$

$$X_2 = T_2$$

$$X_3 = T_3$$

$$U = Q_{in}$$

Conductancia (inversa de resistencias térmicas)

$$G_m$$

$$C_1 \dot{T}_1 = G_6(T_2 - T_1) + G_7(T_3 - T_1) \quad (1)$$

$$C_2 \dot{T}_2 = G_1(T_a - T_2) + G_6(T_1 - T_2) + G_3(T_3 - T_2) + Q_{in} \quad (2)$$

$$C_3 \dot{T}_3 = G_2(T_a - T_3) + G_3(T_2 - T_3)$$

$$\begin{bmatrix} \dot{T}_1 \\ \dot{T}_2 \\ \dot{T}_3 \end{bmatrix} = \begin{bmatrix} \frac{-G_6 + G_7}{C_1} & \frac{G_6}{C_1} & 0 \\ \frac{G_6}{C_2} & \frac{-G_1 + G_6 + G_3}{C_2} & \frac{G_3}{C_2} \\ 0 & \frac{G_3}{C_3} & \frac{-G_2 + G_3}{C_3} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{C_2} Q_{in} \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{G_7}{C_1} T_a \\ \frac{G_1}{C_2} T_a \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 \end{bmatrix}$$

$$C_1 = 3 \times 10^6 \text{ J/K}; C_2 = 2 \times 10^6 \text{ J/K}$$

$$C_3 = 1 \times 10^6 \text{ J/K}; G_1 = 200 \text{ W/K}$$

$$G_2 = 150 \text{ W/K}; G_3 = 250 \text{ W/K}$$

$$G_6 = 180 \text{ W/K}; G_7 = 100 \text{ W/K}$$

$$\begin{bmatrix} -9,33 \times 10^{-5} & 6 \times 10^{-5} & 0 \\ 9 \times 10^{-5} & -3,15 \times 10^{-4} & 1,25 \times 10^{-4} \\ 0 & 2,5 \times 10^{-4} & -4 \times 10^{-4} \end{bmatrix} \begin{bmatrix} 0 \\ 5 \times 10^{-7} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix}$$

$$\begin{bmatrix} 3,33 \times 10^{-5} \\ 1 \times 10^{-4} \\ 0 \end{bmatrix}$$

$$F(s) = \frac{5 \times 10^{-7} s^2 + 2,45 \times 10^{-10} s + 1,8 \times 10^{-14}}{s^3 + 0,000805 s^2 + 1,537 \times 10^{-7} s + 6,368 \times 10^{-12}}$$

$$t_{sla} = ?$$

$$t_{sd} = \frac{900}{\dots} s$$

$$\zeta d = 0,9$$

$$W_{nd} = 0,0049$$

$$p(s) = s^5 + 0,142s^4 + 0,00714s^3 + 0,000144s^2 + 9,24e^{-7}s + 2,14e^{-9}$$

$$\phi \hat{A} = [\hat{A}^5 + 0,142\hat{A}^4 + 0,00714\hat{A}^3 + 0,000144\hat{A}^2 + 9,24e^{-7}\hat{A} + 2,14e^{-9}I]$$

$$\hat{A} = \begin{bmatrix} -0,0001 & 0,0001 & 0 & 0 & 0 \\ 0,0001 & -0,0003 & 0,0001 & 0 & 0 \\ 0 & 0,0003 & -0,0004 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 \end{bmatrix}$$

$$\hat{S} = \begin{bmatrix} 0 & 3e^{-11} & -1,21e^{-14} & 5,17e^{-18} & -2,485e^{-21} \\ 5e^{-7} & -1,57e^{-10} & 6,79e^{-14} & -3,36e^{-17} & 1,76e^{-20} \\ 0 & 1,25e^{-10} & -8,93e^{-14} & 5,27e^{-17} & -2,951e^{-20} \\ 0 & 0 & -5e^{-7} & 1,57e^{-10} & -6,79e^{-14} \\ 0 & -5e^{-7} & 1,57e^{-10} & -6,79e^{-14} & 3,36e^{-17} \end{bmatrix}$$

$$K = [0 \ 0 \ 0 \ 0 \ 1]^{-1} \hat{S} \phi \hat{A}$$

$$\phi \hat{A} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ -0,0006 & -0,1415 & -0,0009 & 0 & 0,0009 \\ 0 & -0,0009 & 0 & 0 & 0 \end{bmatrix}$$

$$K = \begin{bmatrix} K_1 & K_2 & K_3 & K_4 & K_{i2} \\ 273,31e^{10} & 282835,54 & -28,93e^{10} & -118940,53 & 15,67e^8 \end{bmatrix}$$